

Lab 1

Diode

**Electronics and Photonics Post-undergraduate Second
Degree Program**

National Yang Ming Chiao Tung University

Outline



Forward Bias: Knee Voltage



Reverse Bias



Half-wave Rectifier



(O) Full-wave Rectifier

Forward Bias: Knee Voltage

Objective

- Observe and verify forward-biased diode conduction characteristics.
- Measure I-V relationship to identify the Knee Voltage.
- Understand non-linear diode behavior and its fundamental applications.

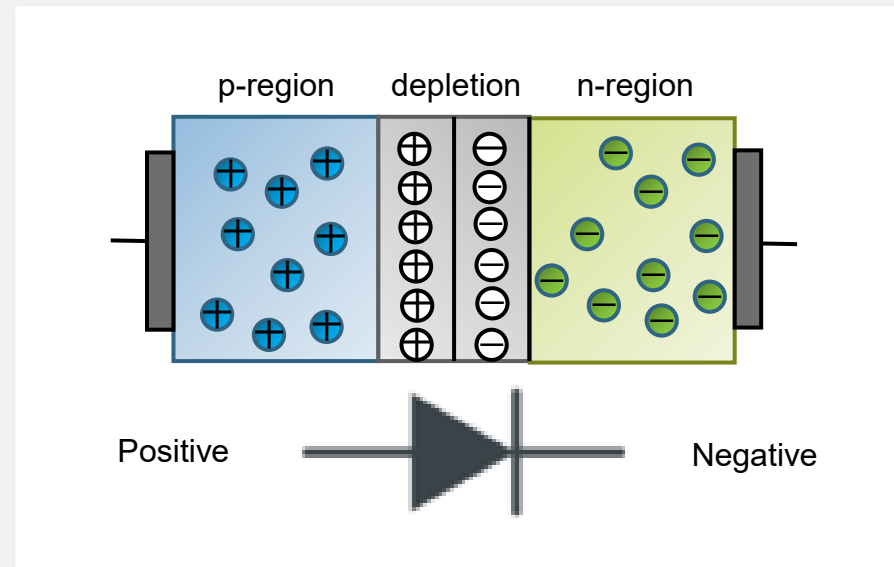


Fig.1.1 PN junction diode

Theory Overview

Diode Basics: A semiconductor device allowing current flow primarily in one direction, formed by a PN junction.

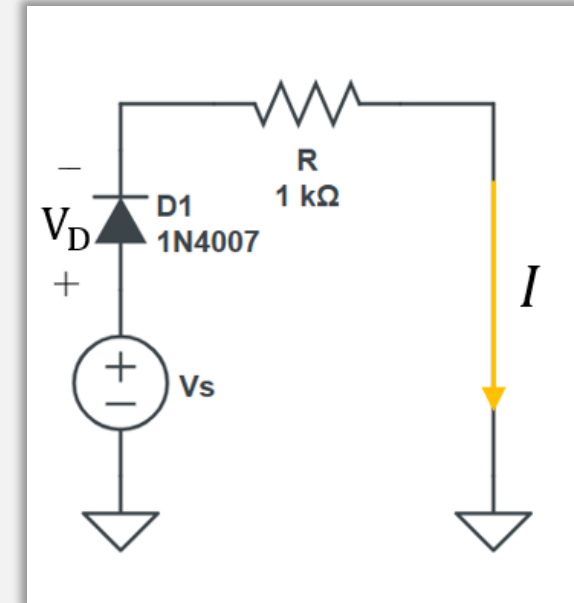
Forward Bias: Positive voltage applied to the P-region, negative to the N-region. This narrows the depletion region.

Knee Voltage: The threshold voltage where the depletion region becomes narrow enough for significant current to flow.

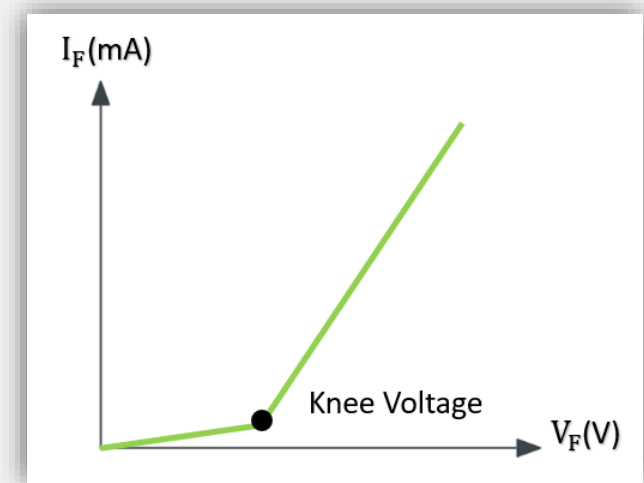
Conduction Behavior: Below the Knee Voltage, the diode exhibits high impedance with minimal leakage current. Above this threshold, its resistance drops sharply, causing current to increase exponentially.

Procedure

- Assemble a forward-biased diode circuit..
- Vary the applied voltage in 0.1V increments while measuring corresponding current and voltage.
- Plot the diode's Current-Voltage (I-V) characteristic curve.
- Identify the Knee Voltage from the plotted curve.



1	0.1		
2	0.2		
...			



Equipment List

Category	Specifications
DMM	GW Instek GDM-532
Power Supply	GW Instek GPE-3323

Component List

Category	Specifications
Breadboard	Sockets: 830 (630 tie-point terminals + 200 power rails)
Wire	22AWG (14 lengths × 5 pcs for each)
Resistance	1 k Ω (1/4W, $\pm 0.5\%$)
Diode	1N4007

Question

1. Based on your plotted I-V characteristic curve, what is the estimated knee voltage of the silicon diode you used in this experiment? How does this value compare to the theoretical knee voltage for silicon diodes? Please explain any observed discrepancies or deviations from the theoretical value?
1. Explain the physical reasons why the current flowing through the diode remains almost zero when the forward bias voltage is below the knee voltage. Conversely, why does the current increase sharply and exponentially once the forward bias voltage exceeds the knee voltage? Relate your explanation to the behavior of the depletion region and majority carriers.
1. What is the primary role of the series current-limiting resistor in this experiment's circuit? Describe what might happen to the diode and/or the power supply if this resistor were omitted from the circuit during forward bias operation.

Reverse Bias

Objective

- Investigate and characterize standard diode behavior under reverse bias.
- Measure small reverse current and observe its high-resistance state.
- Understand the concept of breakdown voltage and diode limitations under reverse voltage.
- Distinguish standard diodes from Zener diodes designed for stable breakdown.

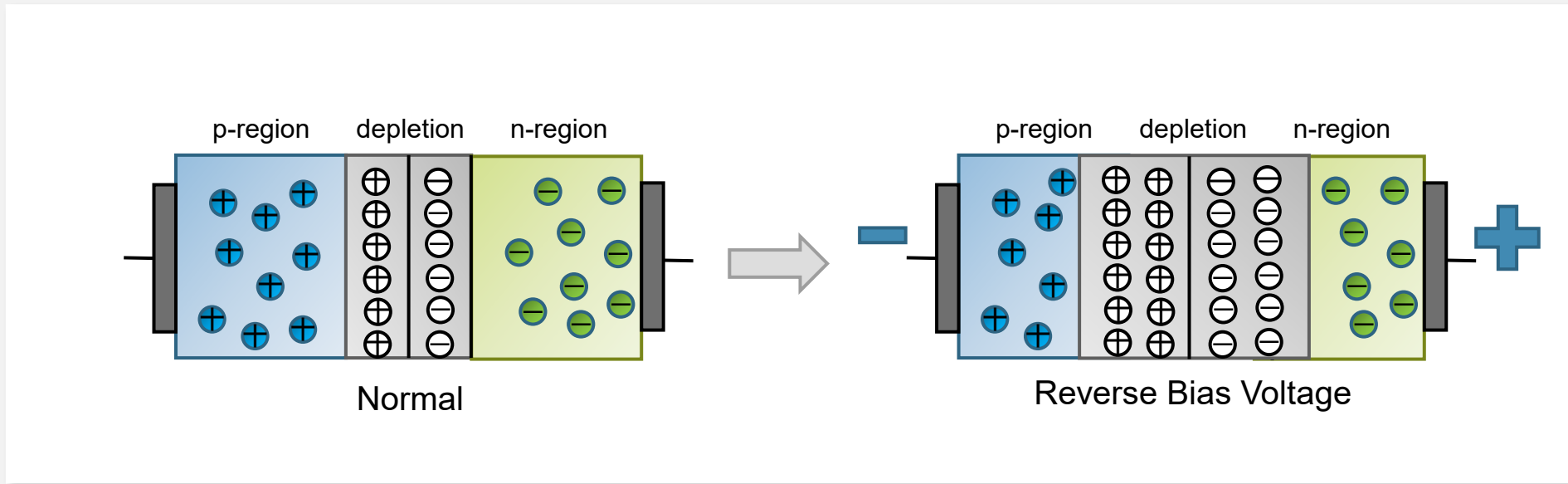


Fig.1.1 PN junction diode

Theory Overview

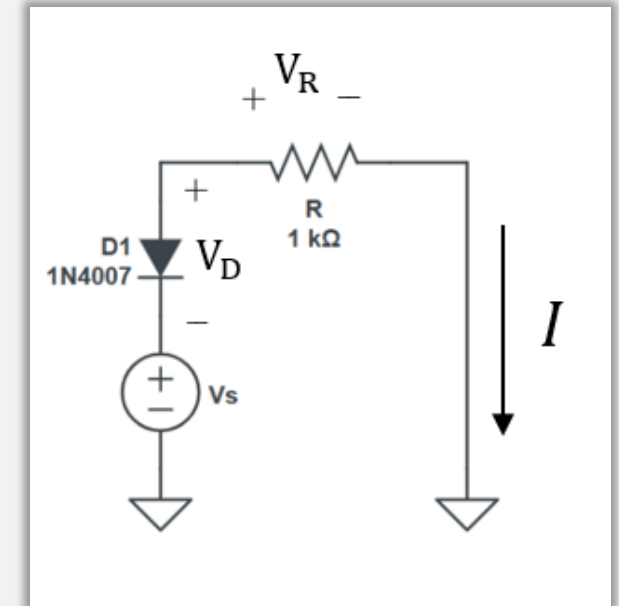
Reverse Bias Fundamentals : Applying reverse bias widens the diode's depletion region, creating high resistance and allowing only a very small, temperature-dependent leakage current.

Breakdown Voltage : A critical reverse voltage causes Zener or avalanche breakdown, leading to a sudden, sharp increase in current.

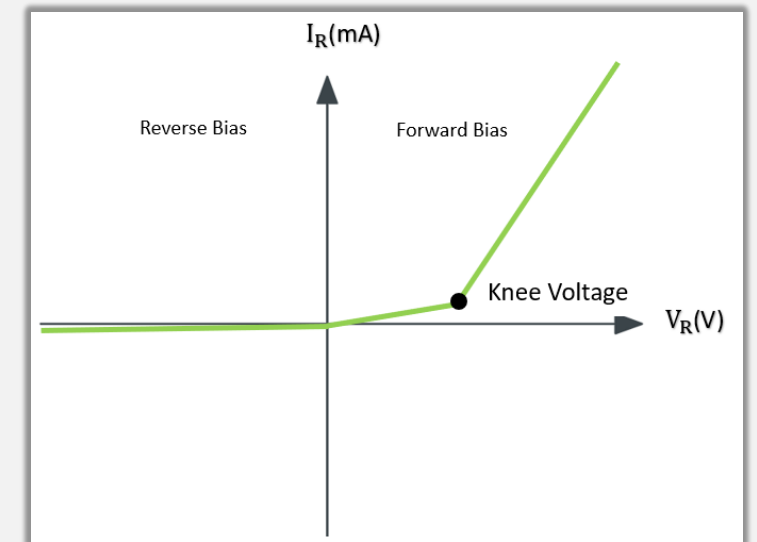
Standard Diode Limitation : Unlike Zener diodes, standard diodes are not designed for stable operation in breakdown; exceeding current limits in this region can cause permanent damage.

Procedure

- Assemble a reverse-biased diode circuit.
- Vary the applied voltage in 1V increments while measuring corresponding current and voltage.
- Plot the diode's Current-Voltage (I-V) characteristic curve.



	V_s (V)	V_D (V)	I (mA)
1	1.0		
2	2.0		
...			



Equipment List

Category	Specifications
DMM	GW Instek GDM-532
Power Supply	GW Instek GPE-3323

Component List

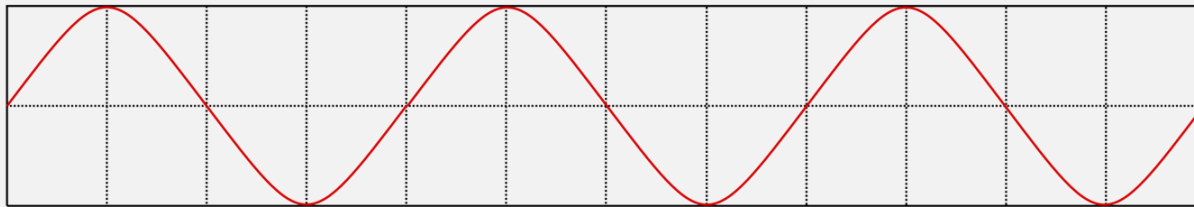
Category	Specifications
Breadboard	Sockets: 830 (630 tie-point terminals + 200 power rails)
Wire	22AWG (14 lengths × 5 pcs for each)
Resistance	1 k Ω (1/4W, $\pm 0.5\%$)
Diode	1N4007

Question

1. Based on your plotted I-V characteristic curve, what is the estimated breakdown voltage (V_{BR}) of the silicon diode used in this experiment? How does this value compare to the diode's datasheet specification (if available)?
2. Describe the phenomenon of "breakdown" in a diode under reverse bias. What are the two main mechanisms (avalanche and Zener) that can cause breakdown, and how do they differ fundamentally? Why is operating a standard diode in this region generally discouraged?
3. What is the role of the series resistor in the reverse bias Zener diode circuit, especially when the diode reaches its breakdown voltage? What would be the consequences if this resistor were omitted during the breakdown test?

Half-Wave Rectifier

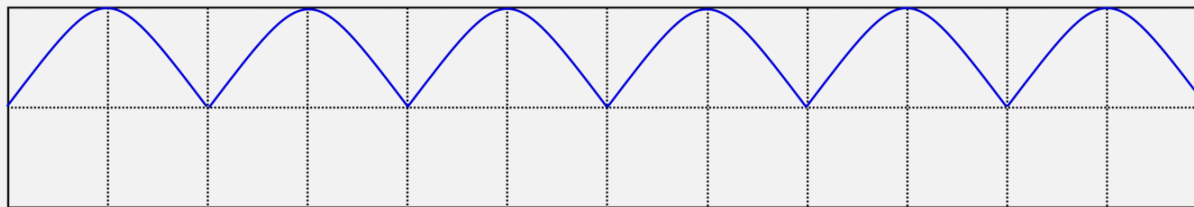
Introduction to Rectifiers



Sinusoidal Signal



Half-wave Rectification



Full-wave Rectification

<https://zh.wikipedia.org/zh-tw/%E6%95%B4%E6%B5%81%E5%99%A8>

- ✦ Rectification converts an **alternating current (AC) waveform** into a **direct current (DC) waveform** with a single polarity, akin to "setting something straight."
- ✦ **DC doesn't require a constant value** (e.g., a battery); it simply means the polarity remains unchanged, with varying amplitudes called pulsating DC.

Half-wave Rectifier

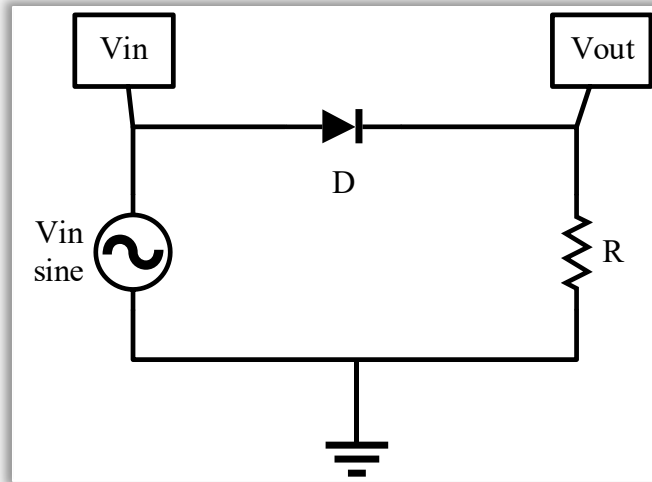


Figure 1. Basic AC diode-resistor circuit.

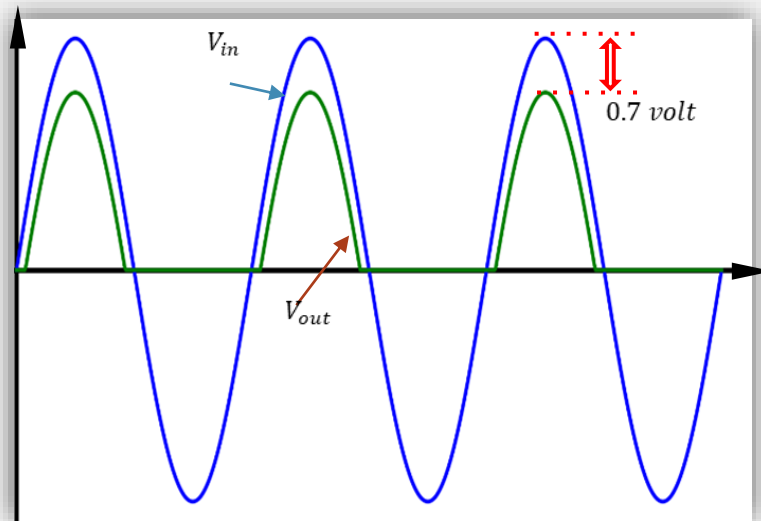


Figure 2. Half-wave rectification waveforms including forward diode drop.

- During the **positive half-cycle** of the input signal, the diode is **forward-biased** and behaves like a **closed switch**, allowing current to flow through the load resistor, across which the input voltage drops.
- During the **negative half-cycle**, the diode becomes **reverse-biased**, acting like an **open switch**. As a result, current stops flowing and no voltage appears across the resistor.

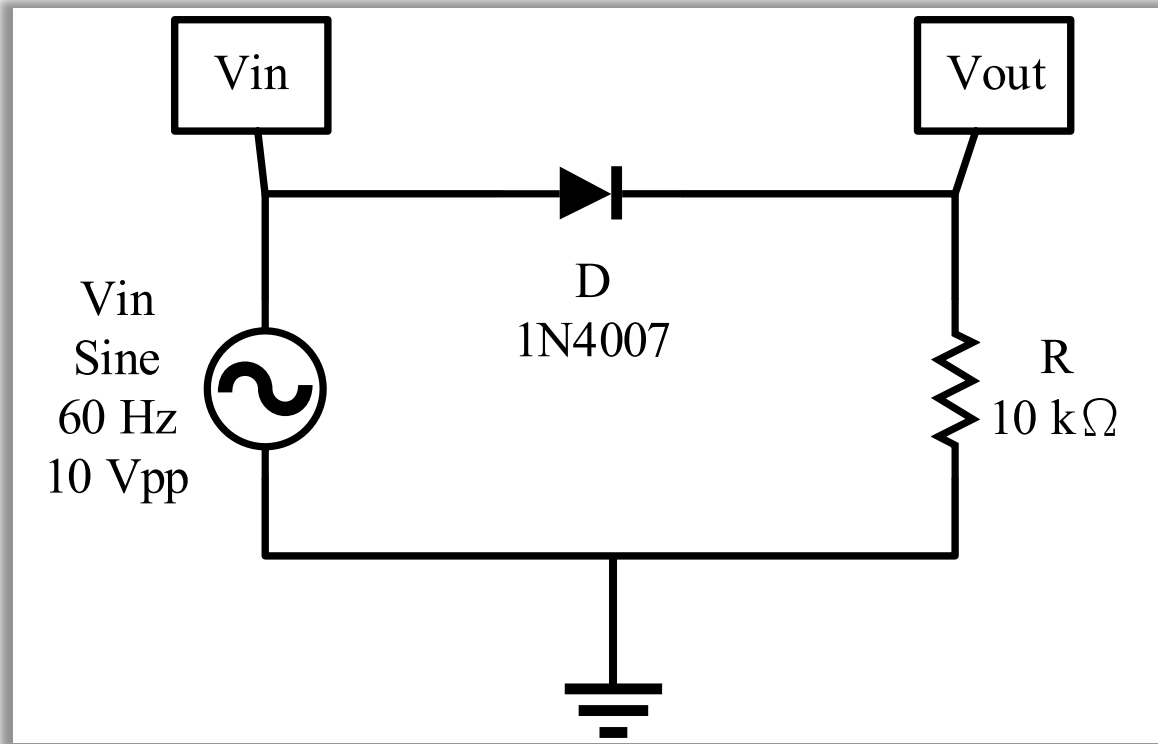
- Arithmetic mean voltage

$$V_{out(avg)} \approx \frac{V_m}{\pi}$$

- **Root Mean Square (RMS) Output Voltage**

$$V_{out(rms)} \approx \frac{V_m}{2}$$

Experiment Procedure - Half-wave Rectifier



Circuit Setup

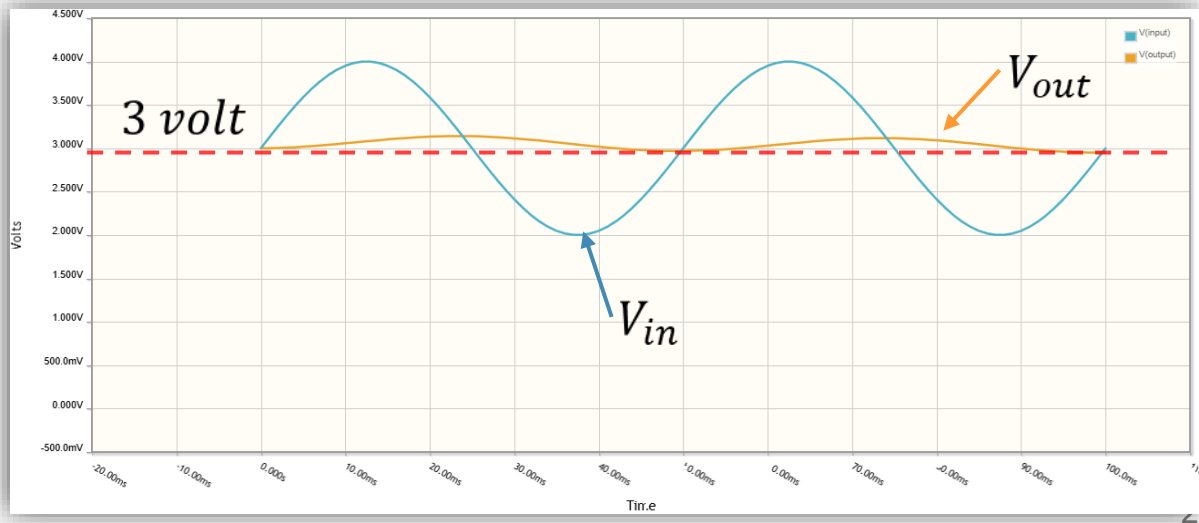
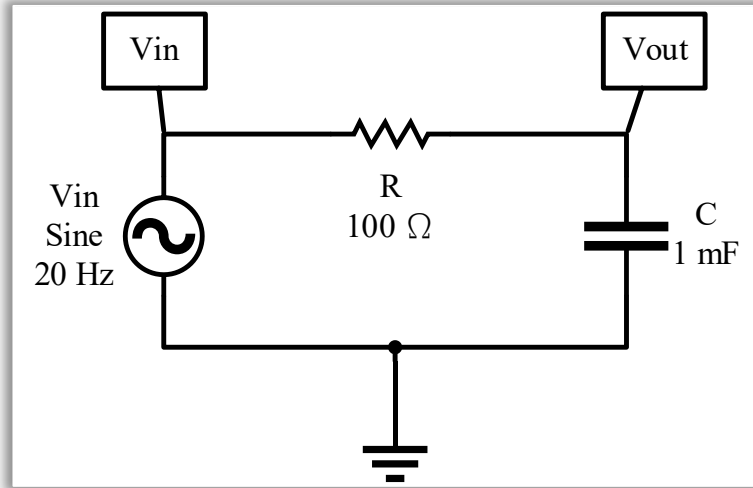
- Connect the circuit according to the diagram on the right.

Measurement

Experiment setup	Wave pattern
Ex.	Ex.

	V_{peak}	V_{rms}	efficiency
V_{in}			
V_{out}			

RC Circuit Review



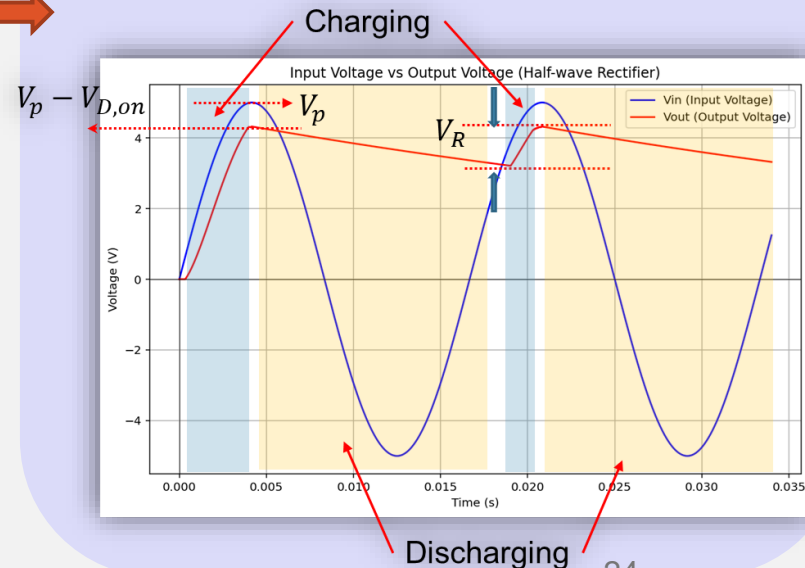
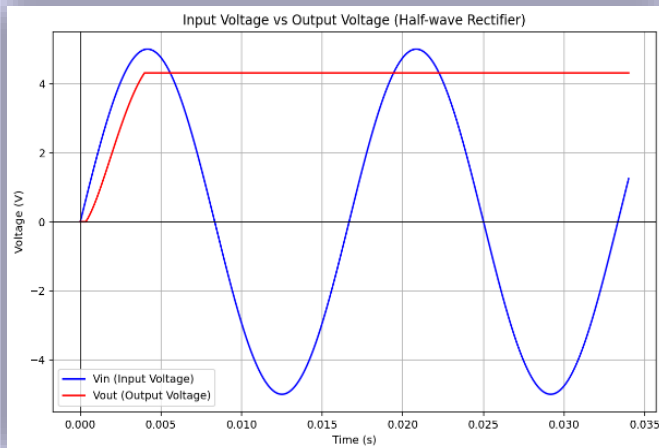
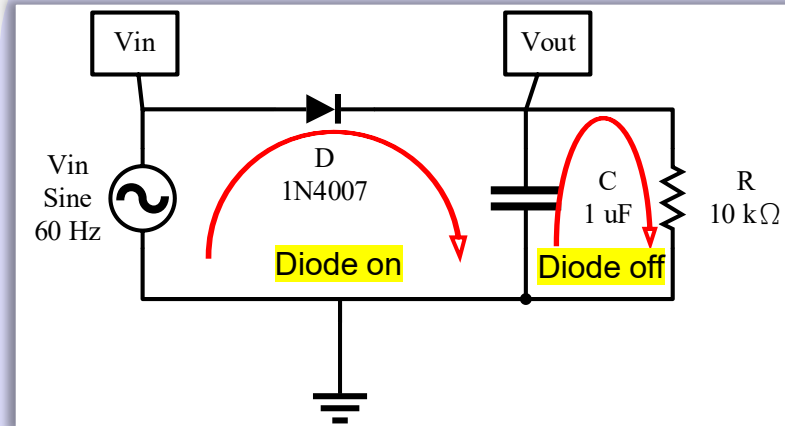
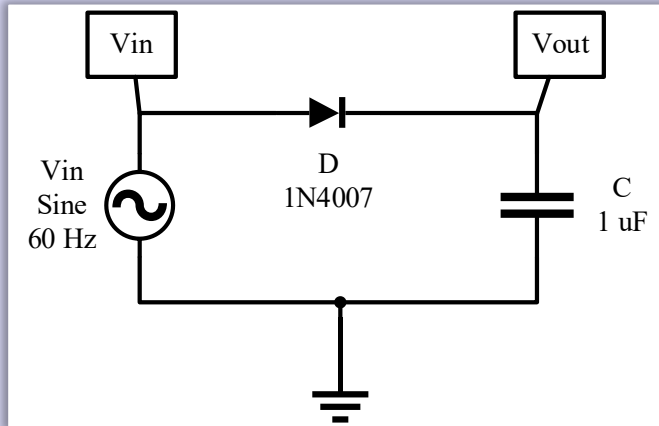
- The RC filter is added to **smooth the output voltage** by **reducing the ripple**, ensuring a more **stable DC** voltage
- The equation below describes the charging and discharging behavior of the capacitor.

$$R \frac{dQ}{dt} + \frac{1}{C} Q = V(t)$$

- The output voltage (V_{out}) across the capacitor is a filtered version of the input, showing **reduced amplitude** due to the **RC time constant**.

$$\tau = RC = 0.1 \text{ s}$$

Half-wave Rectifier with RC filter



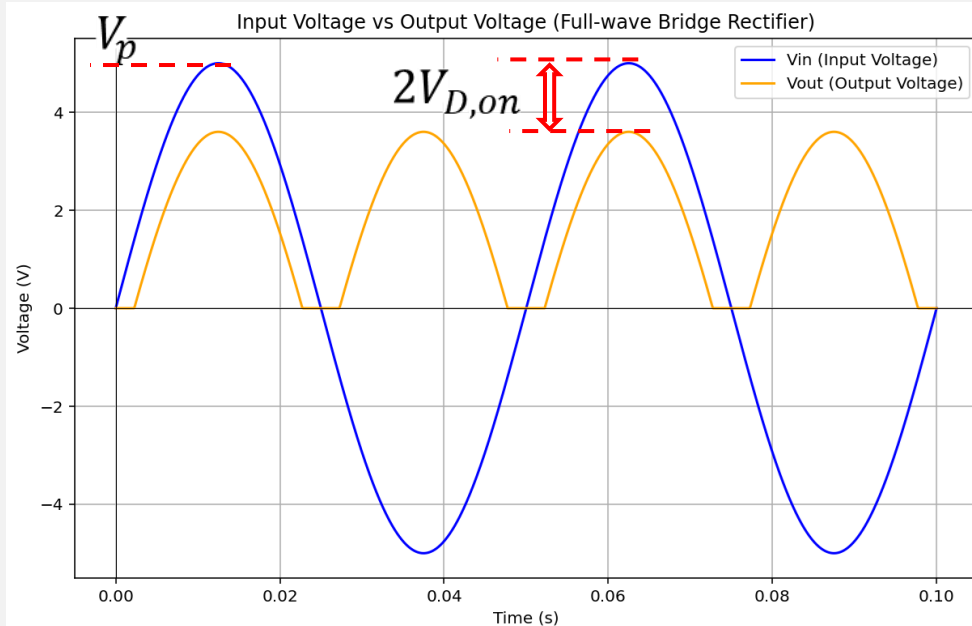
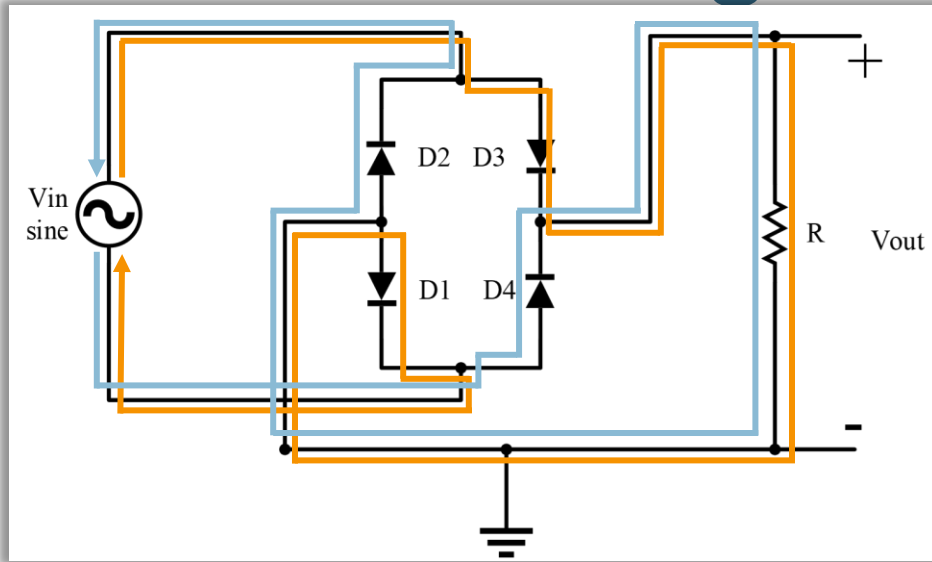
- During the **negative half-cycle**, the diode is reverse-biased (turns "**off**"), and the **capacitor discharges** through the load resistor (R1), **smoothing the output voltage**.

- Assume the frequency of V_{in} is f_{in}

$$V_R = \frac{V_p - V_{D,on}}{R_L} \frac{1}{C f_{in}}$$

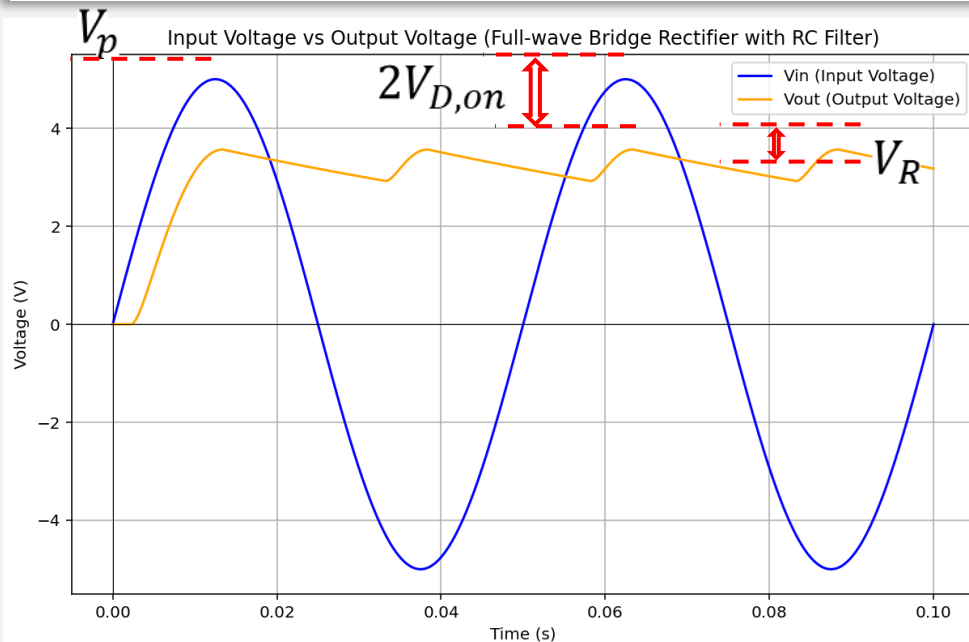
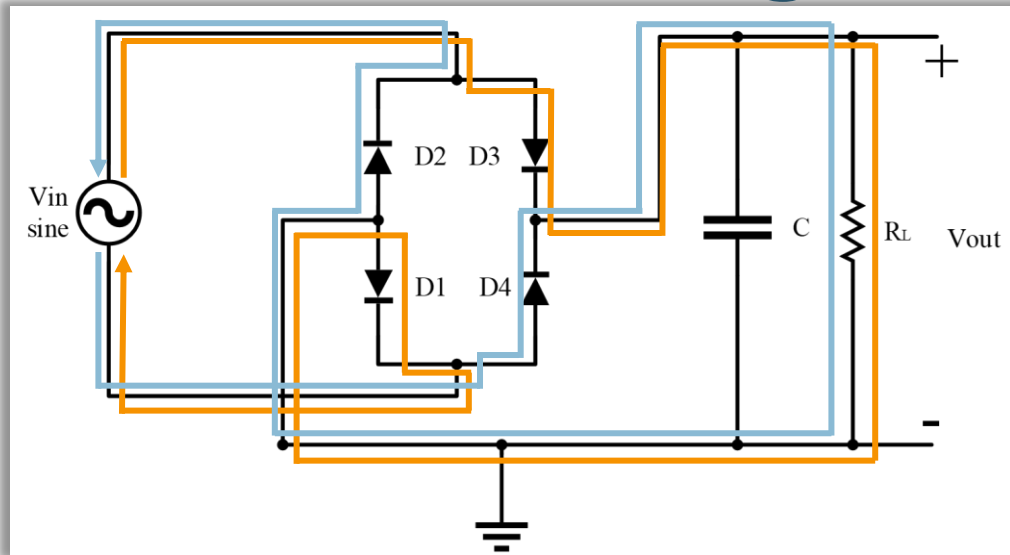
Full-Wave Rectifier

Full-wave Bridge Rectifier without RC filter



- ✦ With D1 and D3 conducting in the positive half-cycle
- And D2 and D4 in the negative half-cycle, directing current through the load resistor (R) in one direction.
- The dead-zone around V_{in} arises because V_{in} must exceed $2V_{D,on}$ to turn on the bridge.

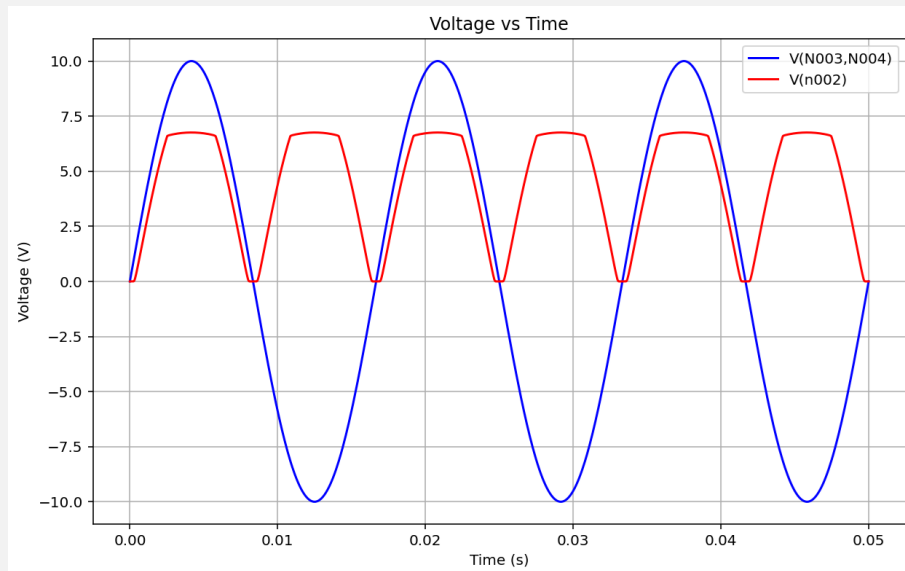
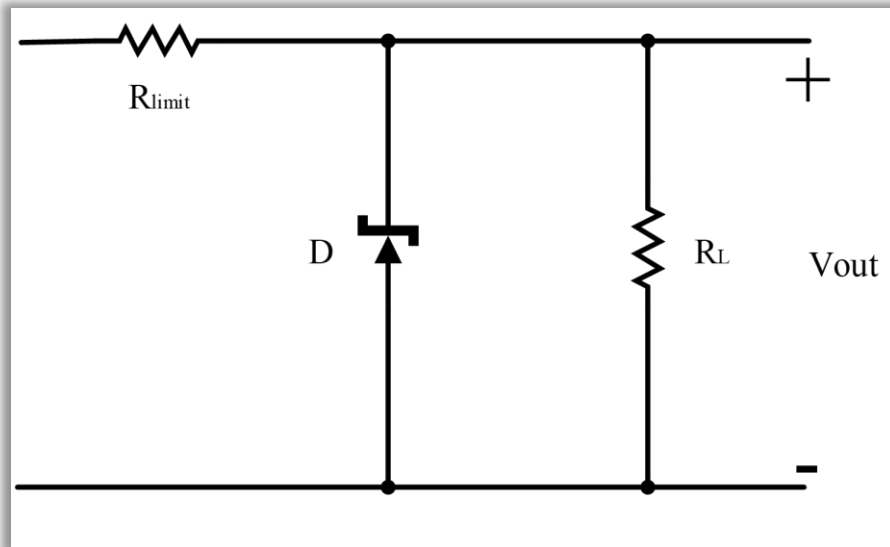
Full-wave Bridge Rectifier with RC filter



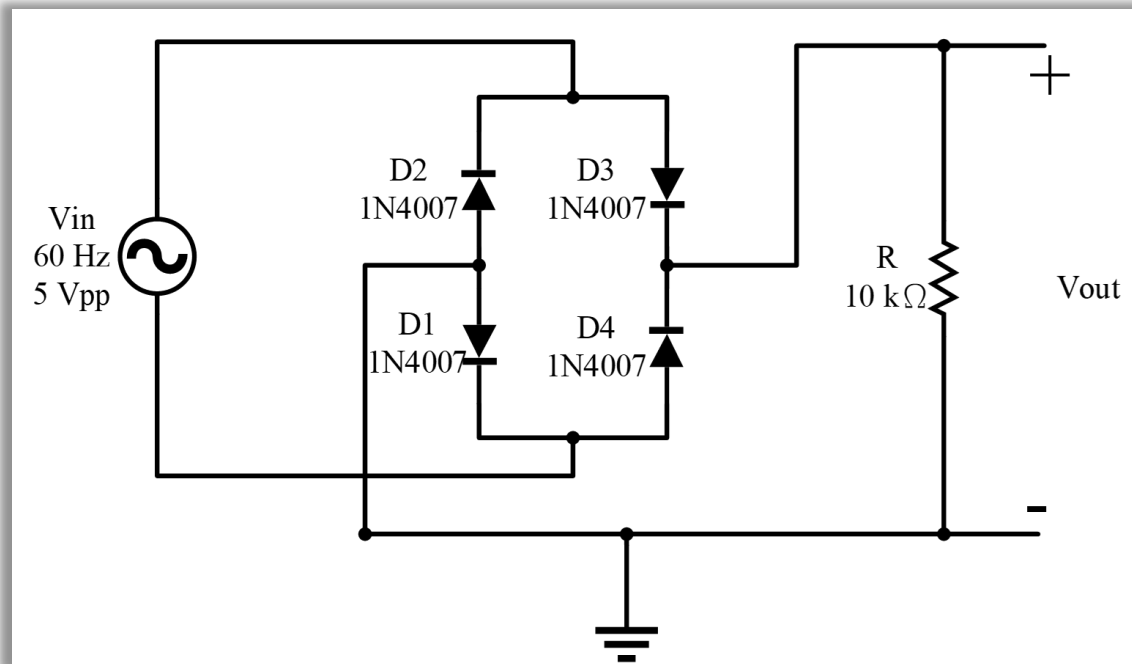
- ✦ The **RC filter**, comprising a capacitor (C) and resistor (R), charges during diode conduction and discharges to smooth the output voltage (V_{out}), significantly **reducing ripple** for a stable DC output
- Since C_1 only gets $\frac{1}{2}$ of period to discharge, ripple voltage is decreased by a **factor of 2**. Also (b) shows that each diode is subjected to approximately one **V_p reverse bias drop** (versus $2V_p$ in half-wave rectifier).

$$V_R \approx \frac{1}{2} \frac{V_p - 2V_{D,on}}{R_L} \frac{1}{C f_{in}}$$

Zener Regulation



Experiment Procedure - Full-wave Bridge Rectifier without RC filter



Circuit Setup

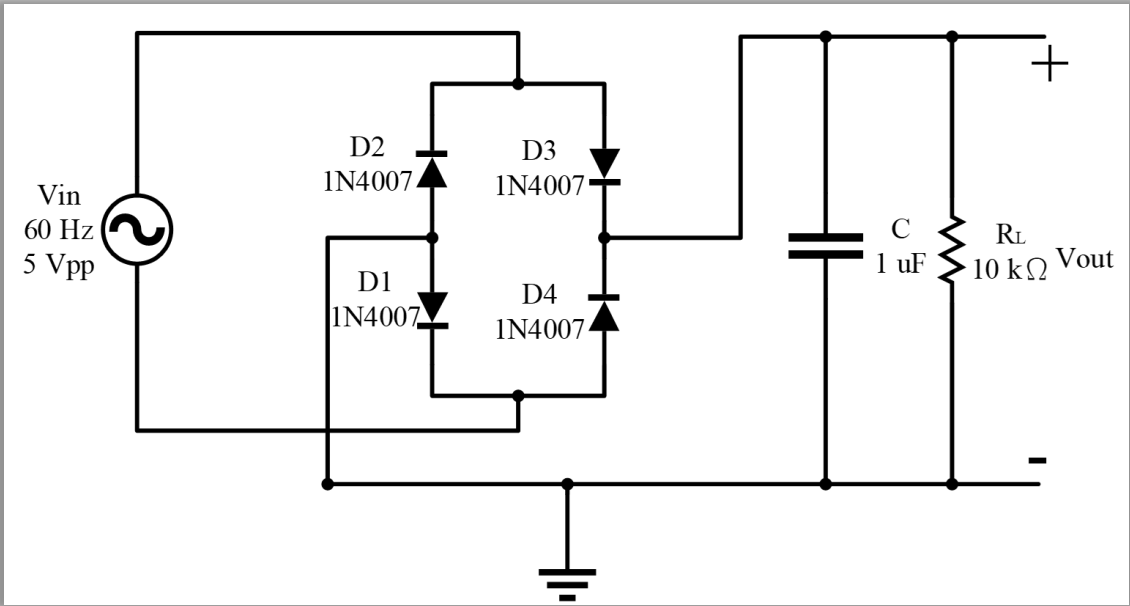
- Connect the circuit according to the diagram on the left.

Measurement

Experiment setup	Wave pattern
Ex.	Ex.

	V_{peak}	V_{rms}	efficiency
V_{in}			
V_{out}			

Experiment Procedure - Full-wave Bridge Rectifier with RC filter



Circuit Setup

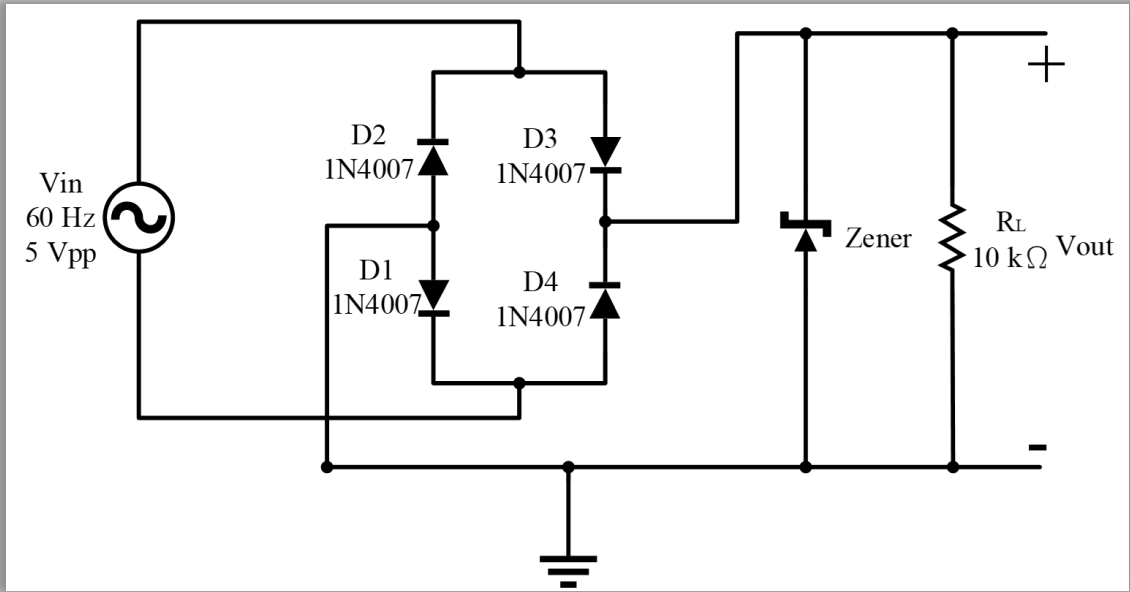
- Connect the circuit according to the diagram on the left.

Measurement

Experiment setup	Wave pattern
Ex.	Ex.

	V_{peak}	V_{rms}	V_R	Ratio (V_R/V_p)
V_{in}				
V_{out}				

Experiment Procedure - Full-wave Bridge Rectifier with RC filter



Circuit Setup

- Connect the circuit according to the diagram on the left.

Measurement

Experiment setup	Wave pattern
Ex.	Ex.

	V_{peak}	V_{drop} (at $V_{in} < V_Z$)
V_{in}		
V_{out}		